

Published Research Summaries

1. Advanced Root Cause Analysis in Complex Mechanical Systems

Author(s): Singh, Y., et al. (2020) **Journal:** Journal of Applied Engineering & Research

Abstract: As mechanical assemblies become increasingly complex, traditional surface-level diagnostic tools fail to identify micro-variations that cascade into systemic failures. This paper introduces a novel framework combining Six Sigma datum tracking with sequential logic analysis to identify implicit root causes in automotive framing stations. By analyzing dimensional variance across 5,000 pilot production units, our methodology reduced fitment deviations by 15%. This approach emphasizes evidence-based deduction over trial-and-error, establishing a rigorous diagnostic protocol applicable to any multi-stage complex system, including computational and logical pipelines.

Note: The rigorous academic standard applied to this publication is the same standard I apply to technical writing and methodology documentation for AI evaluation.